

Table of Contents

MODULE 1: INTRODUCTION TO ARTIFICIAL INTELLIGENCE 1

 1.1 What Is Artificial Intelligence? 1

 1.2 The Foundations of Artificial Intelligence10

 1.3 The History of Artificial Intelligence.....16

 1.4 The State of the Art25

 1.5 Risks and Benefits of AI.....29

 Key Terms Introduced in This Module.....32

 Summary of Module 133

 Review Questions.....34

MODULE 2: INTELLIGENT AGENTS35

 2.1 Agents and Environments.....35

 2.2 Good Behavior: The Concept of Rationality.....38

 2.3 How Agents Should Act: PEAS and the Nature of Environments.....41

 2.4 The Structure of Intelligent Agents45

 2.5 Summary and Comparison of Agent Types.....57

 Key Terms Introduced in This Module.....58

 Summary of Module 258

 Review Questions.....59

MODULE 3: PROBLEM SOLVING60

 3.1 Problem-Solving Agents.....60

 3.2 Search Problems and Solutions: Formulating Problems.....61

 3.3 Well-Defined Problems and Example Problems.....63

 3.4 Measuring Problem-Solving Performance65

 3.5 Search Algorithms66

 3.6 Uninformed Search Strategies.....67

 3.7 Informed (Heuristic) Search Strategies.....73

 3.8 Heuristic Functions.....79

3.9 Constraint Satisfaction Problems (CSPs)	82
Key Terms Introduced in This Module.....	86
Summary of Module 3	87
Review Questions.....	88
MODULE 4: KNOWLEDGE REPRESENTATION AND REASONING.....	85
4.1 Knowledge-Based Agents.....	85
4.2 The Wumpus World.....	87
4.3 Logic: General Concepts.....	89
4.4 Propositional Logic	90
4.5 First-Order Logic.....	92
4.6 Inference in First-Order Logic.....	95
Key Terms Introduced in This Module.....	104
Summary of Module 4	104
Review Questions.....	106
MODULE 5: PLANNING.....	107
5.1 Definition of Classical Planning	107
5.2 Algorithms for Classical Planning.....	110
5.3 Heuristics for Planning.....	114
5.4 Planning and Acting in Nondeterministic Domains	117
5.5 Time, Schedules, and Resources.....	120
5.6 Analysis of Planning Approaches.....	122
Key Terms Introduced in This Module.....	123
Summary of Module 5	124
Review Questions.....	125